Allen Choi

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EDUCATION

California Polytechnic State University **San Luis Obispo, CA**
Master's + Bachelor of Science in Statistics, Minor in Data Science Expected June 2026

- **Thesis**: *Optimal k in Unsupervised Clustering Algorithms (In Progress)* - Advisor: Dr. Kelly Bodwin
- **Undergraduate GPA:** 3.75 / 4.00
- **Relevant Coursework:** Probability Theory, Estimation and Sampling Theory, Mathematical Statistics, Mathematical Foundations of Data Science

RESEARCH EXPERIENCE

Undergraduate Research Assistant **Sept 2024 - Present**
California Polytechnic State University, San Luis Obispo

- Working under Dr. Trevor Ruiz at Cal Poly to process/clean spatial and time series sensor data using R
- Collaborated with a cross-disciplinary team including professors from the biology department to explore and model factors affecting rattlesnake body temperatures across the US

Undergraduate ISI-BUDS (SIBS) Researcher **June 2024 - August 2024**
University of California, Irvine

- Participated in the ISI-BUDS research program, which is one of the Summer Institute in Biostatistics (SIBS) programs funded by the National Institutes of Health (NIH)
- Hands-on experience through biostatistics related project while learning modern statistical methodology, sharing insights at end-of-program presentation with a team of two other students

PROFESSIONAL EXPERIENCE

Data Analyst Intern **June 2025 – Aug 2025**
GALLO Modesto, CA

- Developed and deployed a comprehensive data quality monitoring tool utilizing Python, SQL, and statistical analysis, streamlining the resolution of data quality issues for GALLO's data governance team
- Ensured successful tool adoption through regular stakeholder communication, Git-based version control, and hands-on training delivery within an **Agile**/**Scrum** development framework

Statistics Tutor **Oct 2023 – June 2025**
California Polytechnic State University, San Luis Obispo San Luis Obispo, CA

- Facilitated tutoring as part of Cal Poly's Learning Support Center, supporting up to 10 different statistics courses
- Provided individualized tutoring in statistics fundamentals and software, helping students significantly improve understanding and boost performance in statistics courses

Instructional Student Assistant **Winter 2024, Present**
California Polytechnic State University, San Luis Obispo San Luis Obispo, CA

- Worked as a student assistant for a statistics and computer science class respectively to efficiently grade assignments and provide feedback to students

PALIiSaDS Mentor **Sept 2025 - Present**
California Polytechnic State University, San Luis Obispo

- Serving as a mentor to two undergraduate students as part of the PALIiSaDS program to provide general academic support, assist in preparation of graduate school applications, and regularly meet to ensure their academic success

RELEVANT PROJECTS

Facial Recognition Model (Python - TensorFlow/PyTorch)

- Developed and trained Multi-Layer Perceptron (MLP) and Convolutional Neural Network (CNN) models in TensorFlow/Keras, also utilizing transfer learning with cutting-edge Residual Network (ResNet-50) in PyTorch
- Attempted to address biases in AI predictions through constant model evaluation and refinement, achieving notable improvements in accuracy

Cancer Prediction Model (Python - scikit-learn)

- Cleaned and preprocessed census data of 400k+ total participants' demographic information
- Utilized scikit-learn to construct and fine-tune a K-Nearest Neighbors, Random Forest, and an ensemble model to classify cancer history in people, optimizing hyperparameters to maximize recall scores

SCHOLARSHIPS

(2023 - 2025) PALIiSaDS Scholar Scholarship ($10,000 / year)

- The Pacific Alliance for Low-income Inclusion in Statistics and Data Science program is funded by the National Science Foundation to support low-income and high-achieving undergraduate students

(2024) Herbert E. Collins Scholarship
(2024) JSM Travel Funding

SKILLS

Programming Languages: R, Python, SQL, Java, SAS
Software: GitHub, Tableau, RStudio, VSCode
Languages: English (Fluent), Korean (Fluent)